

3.9 Derivatives of Exponential and Logarithmic Functions

Recall: In section 3.6 we established the following theorem.

Theorem: Suppose that b is a positive real number constant. Then,

$$\frac{d}{dx}(b^x) = b^x \ln(b)$$

Note: This derivative formula agreed with the fact that $\frac{d}{dx}(e^x) = e^x$.

Example 1: Find the first derivative of $y = 3^{\tan(x)}$.

Since we have already seen how to take the derivative of exponential functions we turn our attention to logarithmic functions.

Motivation: We would like to find a formula for the derivative of $y = \log_b(x)$ where $b > 0$ and $x > 0$.

Theorem: Suppose that b is a positive real number constant. Then,

Note:

Example 2: Answer each of the following.

a) Find the first derivative of $y = \ln(\sin(x))$.

b) Find the first derivative of $y = \log_{10}(2x + \cos(x))$.

c) Find the first derivative of $y = \ln|x|$.

Definition: Logarithmic Differentiation is a process of differentiating a function by first taking the logarithm of both sides and then using implicit differentiation to find the derivative.

Note: Properties of logarithms will allow us to make complicated functions much easier to differentiate.

Properties of Logarithms:

Product Rule: $\log_a(xy) =$

Quotient Rule: $\log_a\left(\frac{x}{y}\right) =$

Power Rule: $\log_a(x^n) =$

Note: The necessary restrictions on x , y , a , & n have been omitted here.

Example 3: Use logarithmic differentiation to find the first derivative of $y = \frac{\sqrt{x^2+1}}{(3x+2)^5}$.

In the above example the process of logarithmic differentiation was not needed. The next example shows a time when this process is imperative.

Example 4: Find the first derivative of $y = x^x$, assuming $x > 0$.

Example 5: Use logarithmic differentiation to prove the power rule.

The following is an interesting fact which is oftentimes quite useful.

$$\textbf{Fact: } \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

Explanation (if time permits):